Examiner only

4. Some students investigate the superposition of sound waves. Two identical loudspeakers (S_1 and S_2) are connected to the same signal generator and act as coherent sources.

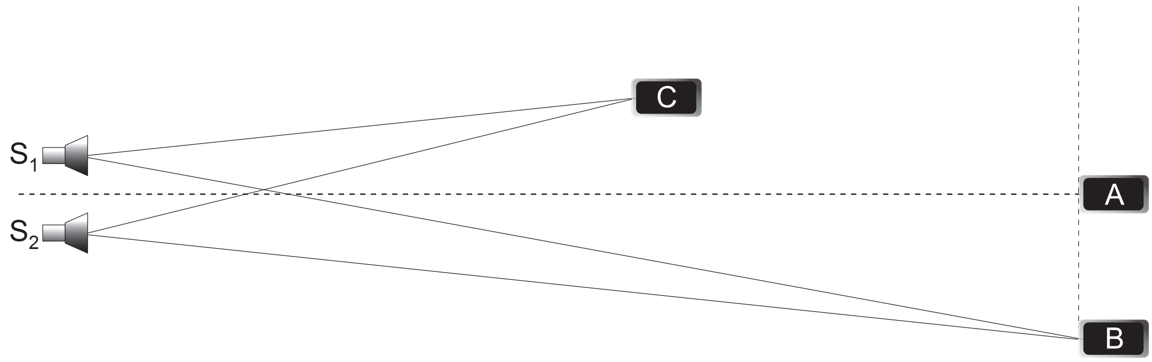

Diagram not drawn to scale

The students use an app on their mobile phones to measure the intensity of the sound signal at different positions in the classroom. Mobile phone A detects a maximum of intensity on the central line. Mobile phone B detects the **first** maximum away from the central maximum.

(a) Explain what is meant by coherent sources. [1]

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(b) (i) The distance S_1B is 8.0 m and the distance S_2B is 7.4 m. State what is meant by the term path difference **and** hence explain why the wavelength of the sound wave must be 0.60 m. [2]

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(ii) Mobile phone C detects a minimum intensity. Determine **two** possible values of the distance ($S_2C - S_1C$). [2]

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Examiner only

(c) (i) State the principle of superposition. [2]

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(ii) Loudspeaker S_2 develops a fault whose only effect is to reduce the amplitude of the sound waves it emits. Explain why the intensities at A and B decrease **and** the intensity at C increases. [3]

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